

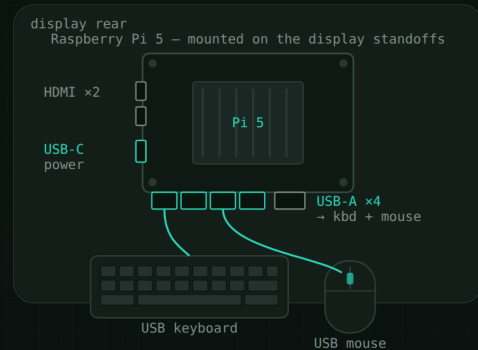
7" TOUCHSCREEN BUILD • QUICK GUIDE

Pi5 DECK

The simplest build: a Raspberry Pi 5 mounted behind the 7" Touch Display 2, with a USB keyboard and mouse. Mount, flash, plug in, boot.

Pi5 DECK // minimal – screen on Pi

7" touch + Pi 5 only • external USB keyboard & mouse • USB-C powered



REAR – Pi 5 stacked behind
≈ 28 mm deep (display + Pi + cooler)

00 Overview

This is the minimal Pi5 DECK: an all-in-one 7" touchscreen computer. No battery, no soldering, no custom wiring — the Pi bolts to the display's built-in standoffs and powers over USB-C. You'll have a working machine in an afternoon.

Subsystem	Specification
Compute	Raspberry Pi 5, 8GB (BCM2712, 4xA76 @ 2.4GHz)
Display	Raspberry Pi Touch Display 2 — 7", 720×1280 native, run landscape 1280×720 , 5-pt touch (DSI)
Input	Any USB keyboard + mouse (plug-and-play)
Power	USB-C PD supply (official 27W recommended)
Cooling	Official Active Cooler — still required on a Pi 5
Storage	microSD (simplest); optional 1TB NVMe
OS	Kali Linux (Pi 5 ARM) or Raspberry Pi OS

WANT THE FULL HANDHELD INSTEAD?

The self-contained cyberdeck — physical keyboard, internal 4× 21700 battery, NVMe, 3D-printed slab — is documented separately in the repo (`Pi5-Handheld-Terminal-Blueprint.md` + wiring + CAD). This guide covers the simple 7" build only.

01 Parts

Required

✓	Item	Note
☐	Raspberry Pi 5 — 8GB	BCM2712 board
☐	Touch Display 2 (7")	720×1280 IPS, 5-pt touch — buy a current revision
☐	Active Cooler	Official Pi 5 cooler, or equivalent
☐	microSD	32GB, A2 / U3 (SanDisk Extreme / Samsung EVO)
☐	USB-C PD supply	Official 27W recommended
☐	USB keyboard + mouse	Any — plug-and-play

INCLUDED WITH THE DISPLAY — the 22-pin DSI ribbon and the GPIO power jumper. Nothing else to buy for the core build.

Optional

✓	Item	Adds
☐	M.2 HAT+ + 1TB NVMe (2242)	Fast storage — see §4
☐	USB-C power bank / UPS HAT	Portability / internal battery
☐	Case or stand	Protection / desk use

02 Build It

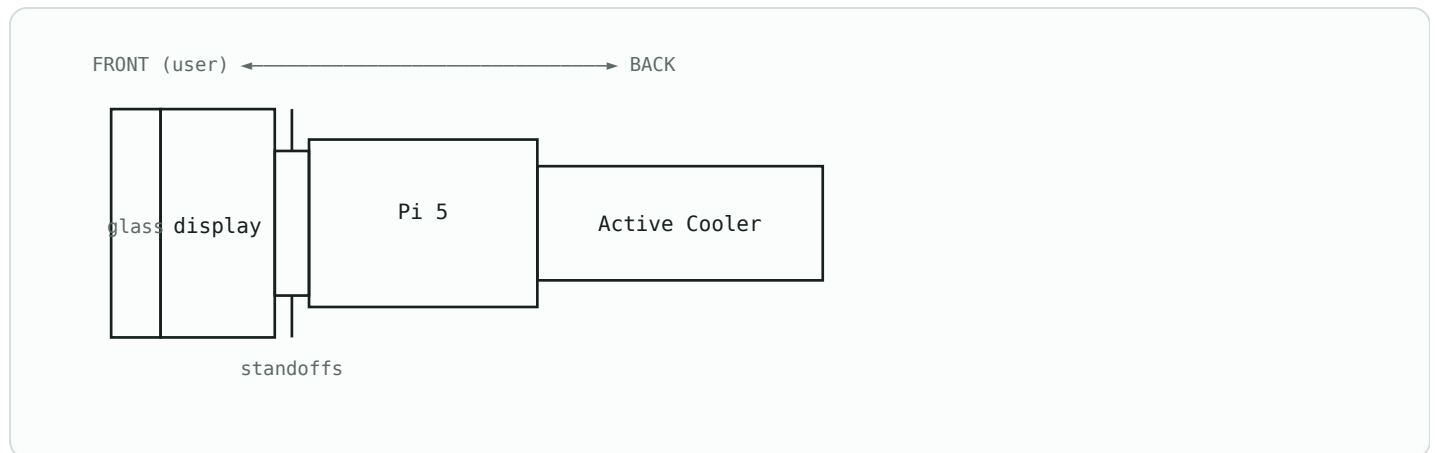


FIG 1 · The stack: Pi 5 bolts to the display's standoffs; the cooler sits on the Pi's SoC facing the back.

1 Mount the Pi

Screw the Pi 5 to the Touch Display 2's pre-installed M2.5 standoffs (58 × 49 mm pattern). Seat the **22-pin DSI ribbon** into **DSI1** (mind the latch), and connect the included jumper: **5V** → **GPIO pin 2**, **GND** → **pin 6** to power the panel.

2 Fit the cooler

Attach the Active Cooler to the SoC and plug its fan into the Pi 5 4-pin fan header. The kernel ramps it automatically. Active cooling is not optional on a Pi 5.

3 Flash the OS

With Raspberry Pi Imager, write Kali (Pi 5 ARM image) or Raspberry Pi OS to the microSD. Set hostname, user, SSH, and Wi-Fi in the advanced options.

4 First boot & configure

Power on with the USB-C supply, then update and set landscape orientation:

```
sudo apt update && sudo apt full-upgrade -y
wlr-randr --output DSI-1 --transform 90 # landscape (Wayland)
```

Add to `/boot/firmware/config.txt`:

```
usb_max_current_enable=1
```

5 Plug in peripherals

Connect the USB keyboard and mouse to any USB-A port — plug-and-play, no drivers. Touch is handled by libinput; recalibrate if the axis is offset after rotating.

DISPLAY REVISION

Pi 5 power-detection can misread **older** Touch Display 2 revisions as a short and refuse to boot. Buy a current revision.

03 Verify

- ☐ Boots to desktop/console on the 7" panel, in landscape
- ☐ Touch tracks correctly (recalibrate if offset)
- ☐ Keyboard and mouse both work
- ☐ Thermals healthy under load

```
vcgencmd measure_temp      # SoC temperature
vcgencmd get_throttled     # 0x0 = healthy
```

DONE

All four boxes ticked? You have a working 7" Pi5 DECK.

04 Optional Add-Ons

Fast storage — NVMe

Add the official M.2 HAT+ and a 1TB **2242** NVMe, then migrate the OS off the microSD with the repo's `nvme-migrate.sh` — run it on the Pi as root. It clones the live system and sets the bootloader to prefer NVMe (it pauses for a `YES` confirmation since it erases the target).

```
chmod +x nvme-migrate.sh
sudo ./nvme-migrate.sh
# after reboot:
findmnt /                # should show /dev/nvme0n1p2
```

Portability

Run from a USB-C PD power bank for cordless use, or add a UPS HAT for an internal battery (this steps toward the full handheld build — see its wiring doc).

Enclosure

Print or buy a case/stand. The full 3D-printed slab enclosure is dimensioned in `Pi5-Deck-CAD-Handoff.md` if you want to grow into the handheld.

05 Appendix

Useful commands

```
vcgencmd measure_temp          # SoC temp
vcgencmd get_throttled          # 0x0 = healthy
wlr-randr --output DSI-1 --transform 90 # landscape
lsusb                           # list USB keyboard/mouse
findmnt /                       # where root is mounted
```

config.txt

```
usb_max_current_enable=1          # allow full peripheral current
# display rotation per Touch Display 2 docs (landscape)
```

Software

- Kali Pi 5 ARM image · kali.org/get-kali
- Raspberry Pi Imager · raspberrypi.com/software
- [rpi-clone](#) (used by `nvme-migrate.sh`) · github.com/billw2/rpi-clone